



Can older workers stay productive? The role of ICT skills and training

Jong-Wha Lee^{a,*}, Do Won Kwak^b, Eunbi Song^c

^a Economics Department, Korea University, Republic of Korea, and Centre for Applied Macroeconomic Analysis (CAMA), Australian National University, Australia

^b Graduate School of International Studies, Korea University, Republic of Korea

^c Department of Economics, School of Business, Monash University Malaysia, and Centre for Applied Macroeconomic Analysis (CAMA), Australian National University, Australia

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ABSTRACT

This paper quantitatively examines the effects of aging on labor productivity using individual worker data in Korea. We find that information and communications technology (ICT) skills and participation in job-related training can help older workers stay productive. The estimation results present that ICT skills, a measure incorporating an individual's proficiency in ICT skills and their utilization in the workplace, has a positive effect on the wages of the older workers aged 50–64 with a high level of education or in a skill-intensive occupation. Job training also has a significant positive effect on the wages of older workers. These results imply that compared to younger workers, well-educated older workers can obtain greater productivity increase through ICT skill attainment and their adequate use, and job-training. The evidence suggests that a productivity decrease in line with the aging process can be mitigated by training aging workers to equip themselves with ICT skills.

1. Introduction

An aging population is an urgent challenge in South Korea (Korea, henceforth). Korea's economy has been experiencing a drastic demographic shift toward an aged society where elderly citizens aged over 65 years will make up a third of its population in 2040—more than double from the 15% that was tabulated in 2019 (Fig. 1). The pace of aging within the workforce is faster than any other economy. The share of workers aged 50–64 years over the working-age population aged 15–64 years began rapidly rising since 1990. It has more than doubled from 16% in 1990 to 33% in 2019. By 2050, it is expected to continue to rise, outpacing Japan and Germany (Fig. 2). With a common belief that older workers tend to be less productive than their younger counterparts, an aging workforce remains as one of the most urgent challenges that the Korean economy will face in the coming years.

An increase in older workers may hinder economic growth if the older workers are less productive than the younger workers. This is from the deterioration of the physical and cognitive abilities, which comes along with age and the low tendency to adapt to new technologies. Maintaining the productivity of older workers is not an easy task. However, if older workers are more educated and has improved their human capital through their work experience, job training, and acquisition of new skills that are adequate for their jobs and tasks after their formal education, they can stay productive.

* Corresponding author at: Economics Department, Korea University, 145 Anam-ro, Seongbuk-gu, Seoul 02841, Republic of Korea.
E-mail addresses: jongwha@korea.ac.kr (J.-W. Lee), dwkwak@korea.ac.kr (D.W. Kwak), eunbi.song@monash.edu (E. Song).

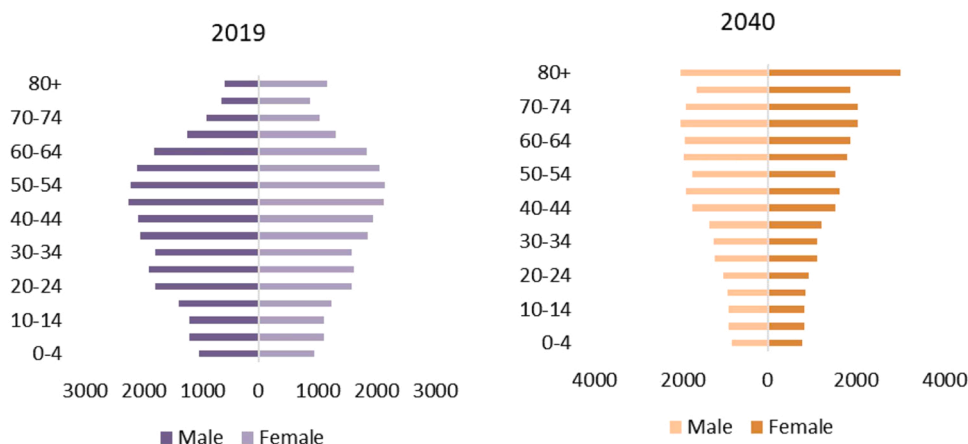


Fig. 1. Population Age Structure in Korea, Selected Years (thousands).

Source: Authors' construction from [United Nations \(2019\)](#), *World Population Prospects, The 2019 Revision*, UN, New York.

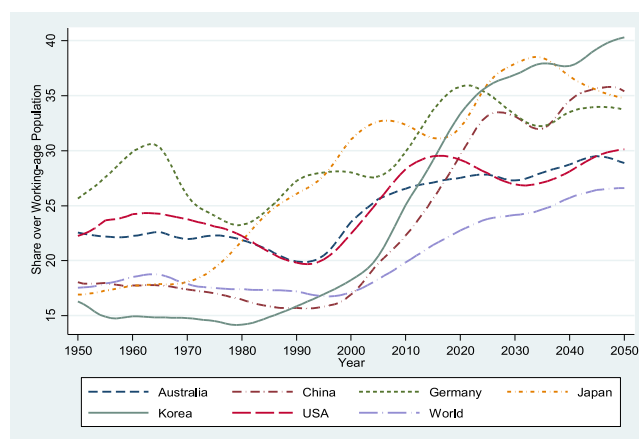


Fig. 2. Share of Workers Aged 50–64 over Working-age Population (16–64 years).

Source: Authors' construction from [United Nations \(2019\)](#)'s *World Population Prospects*.

In Korea, along with the rapid growth of the information and communications technology (ICT) sector and the heavy investment in new technologies in the economy ([Lee, Song, & Kwak, 2020](#)), acquisition of ICT skills has also been active—for instance, through formal education.¹ Korea's ICT use in education essentially began with improving students' computer literacy, followed by continued reinforcement of human and physical infrastructure; based on this approach, there were increased learning opportunities using ICT. For instance, ICT education was activated during the 2001–2005 period. The national curriculum allowed primary and secondary schools to launch an independent course for teaching basic computer and word processing skills for their students. This helped students familiarize themselves with computers and utilize them in their daily lives. For instance, primary students learned how to turn the computers on and off, and how to find information using the internet browser, whereas secondary students were taught how to communicate via the internet and produce documents using word processors.

However, despite a relatively large investment in ICT, there is a question of whether all Korean workers possess adequate cognitive skills required to effectively perform their tasks in the era of information and automation intelligence (AI) technologies. According to the Organisation for Economic Co-Operation and Development (OECD)'s Programme for the International Assessment of Adult Competencies (PIAAC), Korean adults who participated in the PIAAC survey scored below the OECD average in “problem-solving skills in technology-rich environments,” which involved ICT skills ([Korea Research Institute for Vocational Education and Training \(KRI-VET\), 2012](#)), whereas their score for literacy proficiency was above the OECD average. There is also a large differential in ICT skills

¹ The Korean government sequentially developed ICT education. There are four phases of ICT education development according to [Ministry of Education MoE & Korea Education and Research Information Service KERIS, 2014](#): Phase I, Foundation Establishment, over the period of 1996–2000; Phase II, Dissemination and Settlement, from 2001 to 2005; Phase III, Advancement, from 2006 to 2010; Phase IV, Education Converged with Technologies, from 2010 to 2014.

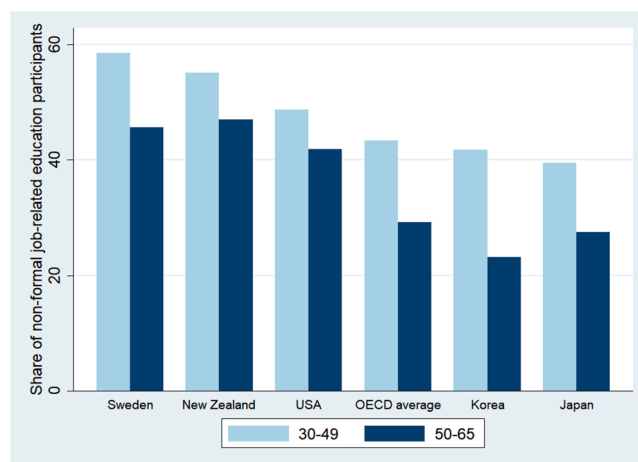


Fig. 3. Participation rates for job training in selected economies. Note: The figures measure a share of those who participated in non-formal job-related education (that is job training) in the previous 12 months at the time of the PIAAC survey among all the respondents. Source: Authors' construction from the OECD (2013a)'s PIAAC Data.

proficiency between older and younger age groups. Another notable feature is that the participation rate in on-the-job training for the older age group in Korea is lower than other OECD countries, including the United States (US), Japan, and Nordic countries (World Bank, 2016). Fig. 3 shows that the job training participation rate for older workers is much lower than the average of the OECD countries, while the rate for prime-age workers (30–49) is only slightly lower than the OECD average.

This study builds on three lines of previous literature, including: (1) those examining the aging effects on productivity; (2) those exploring the role of ICT skills on worker's productivity; and (3) those assessing the role of job training participation in improving worker's productivity. Aging tends to have detrimental effects on productivity as the human capital of an individual declines with age, owing to the deterioration of their physical and cognitive capacities (Truxillo, Cadiz, & Hammer, 2015). Some empirical studies—including Heckman, Lochner, and Todd (2003)—show a hump-shaped age-profile of wages. In addition, adapting to new technology tends to leave them behind. The knowledge and skills acquired by older workers during their formal education are often outdated (ADB, 2018), and old workers tend to have lower adoption rates of new technologies than younger workers (Meyer, 2011).

On the other hand, there are other studies showing that aging does not necessarily accompany productivity declines. If older workers are more educated or continuously develop their human capital after their formal education by adapting to new technologies, having longer work experience, participating in on-the-job training, and equipping themselves with proper skills, their productivity might not necessarily be lower than that of younger workers. Along with a longer lifespan, workers have a greater willingness to work longer, thus having a higher motivation to invest in their health and education (Vogel, Alexander, & Borsch-Supan, 2017; Prettnier et al., 2013). Borsch-Supan and Weiss (2016), using data from a truck assembly plant in Germany, provide evidence that the increase in productivity of individuals up to the age of 60 can be explained by longer work experience and non-cognitive abilities, such as communication skills. Furthermore, firm-level studies across the countries suggest that productivity peak tends to be around the ages 30–45, but the range varies across countries and industries, as summarized in Chomik and Piggott (2019). The Asian Development Bank (2018) reports that there are some sectors where workers may show greater productivity, even at the later stages of their careers. Burtless (2013) further argues that there is little evidence for the negative link between aging and productivity in the US. To sum it up, previous studies on the effect of aging on productivity remain inconclusive.

ICT skills becomes important for individuals to stay productive at the time of rapid advances in ICT and automation technologies (World Bank, 2016). It also indicates that the race between skills and technology could bring about the digital divide between those who can use and complement digital technologies and those who have limited access and skills. Falck, Heimisch-Roecker, & Wiederhold, 2021 report positive returns to ICT skills, measured in problem solving skills in technology-rich environments using the PIAAC data. This reflects a significant role that ICT skills play in determining the labor productivity of individuals. They find that workers with a higher ICT skills proficiency earn 27% more than those with a low proficiency on average. They also report that workers with no computer experience earn around 10% less than those with the most basic computer skills. Hanushek et al. (2015) investigate returns on skills across 23 economies using the PIAAC data and report that ICT skills are positively associated with individual earnings. The returns on ICT skills are estimated around 15.3% in Korea. Lee and Wie (2017) also show significantly positive returns concerning ICT skills proficiency in Japan and Korea.

There exists a considerable number of empirical studies on the effects of job training on wages and the productivity of trainees. According to literature reviews by Leuven (2005), Almeida and Faria (2014), and Cegolon (2015), many studies report positive effects of job training on worker's earnings and productivity. The average wage effects are estimated around 5–10% in the United Kingdom (UK) and the US (Leuven, 2005); around 3.7–21.6% for European countries (Bassanini, Booth, Brunello, De Paola, & Leuven, 2005); and around 2.6–9.0% for Korea (Yoo & Kang, 2010). Contrarily, other studies could not find any significant positive effects (Goux & Maurin, 2000; Pischke, 2001). Some studies have also assessed such effects across age groups. Picchio and Van Ours (2013) and Berg,

Hamman, Piszczek, and Ruhm (2017) document that vocational training, especially on-the-job training, is effective in improving productivity for both younger and older workers, although the effect decreases with age. They are also consistent with the meta-analysis done by Haelermans and Borghans (2012). This differential in training can be attributable to the different motives of younger and older workers, as suggested by Lang (2012) and Dostie and Léger (2014). However, the studies aimed at examining job training for older employees do not always show positive wage effects (Göbel & Zwick, 2013). A recent study by Lee, Han, and Song (2019) using the Korean sample from the PIAAC provides evidence that the wage effects of job-related training are positive and have larger effects for the older workers. Overall, the effects of job training depend on the data, econometric methods, and their contexts.

This paper builds on those three lines of literature on the aforementioned topics and contributes to them by investigating the effects of ICT skills and job training on the productivity of older workers in Korea.² One major contribution that we make to the literature is that we construct an ICT skills measure using an individual's ICT skills proficiency as well as his or her use of ICT skills in the workplace. As noted in OECD (2013a), skill proficiency and skill use are two different concepts. In the previous studies on investigating the effects of ICT skills on wages or productivity considers problem-solving skills in technology-rich environment as a measure for ICT skills. However, it does not consider how such ICT skills are used in the workplace or whether an individual's ICT skills are adequately used in his or her job. In this regard, we attempt to incorporate the ICT skills utilization in the workplace in our new measure for ICT skills, by using information on how frequent and to what extent an individual utilizes his or her ICT skills in the workplace. Hence, constructing a new measure for ICT skills highlights the importance of both attainment and use of ICT skills and allows us to investigate more about ICT skills from both demand and supply sides.

In addition, we adopt a regression imputation method to address a missing data problem in the PIAAC's ICT skills proficiency score. Given the characteristics of the PIAAC survey, individuals who had no experience in computer and/or failed the ICT core/basic assessment do not have scores for ICT skills proficiency. These individuals with missing data are not randomly selected and rather represent individuals who are more likely to have low wage and low ICT skills proficiency score. To control for missing data problems, we use a regression imputation method and high correlation between ICT skill proficiency score and literacy as well as numeracy scores. We apply a regression model where literacy and numeracy scores are used to predict missing ICT skills scores.

The remainder of this paper is organized into five sections. Section 2 provides our empirical specification; Section 3 describes the data; Section 4 examines the wage effects of ICT skills and job training at an individual level, especially for older workers; Section 5 discusses identification strategy; and Section 6 presents the conclusion.

2. Empirical specification and strategy

Over their lifetime, individuals spend a certain period of time in formal schooling and for work after their completion of their formal education. An individual j obtains an educational level of S_j and continues to accumulate his or her own human capital via work experience and job training. We consider that an individual can develop cognitive and non-cognitive capabilities as well as technical skills that can command and complement ICT technologies to perform his or her task at work. Then, human capital attained determines the worker's productivity or wages in the labor markets. We can write the worker's productivity or wages, using the Mincerian wage equation, as follows (Mincer, 1974; Acemoglu, 2009):

$$\log W_j = \alpha + \beta_G \text{Gender}_j + \gamma_s \text{School}_j + \gamma_e \text{exp}_j + \gamma_{e2} \text{exp}_j^2 + \gamma_{ICT} \text{ICT skills}_j + \gamma_T \text{Training}_j + \varepsilon_j, \quad (1)$$

where $\log W_j$ is the logarithm of individual j 's wage that captures the worker's productivity. Wages are determined by the individual's educational level (School_j), work experience (exp_j), ICT skills (ICT skills_j), and job training (Training_j). Gender_j is a female indicator representing whether a worker is female.³

As for ICT skills (ICT skills_j), we construct a composite measure incorporating both ICT skills proficiency of an individual, measured in problem solving skills in technology-rich environments, and his or her use of ICT skills in the workplace, measured in an index of use of skills at work in the PIAAC data. Section 3 explains the construction of this new ICT skills measure. As for job training (Training_j), we use an indicator of job training participation presenting whether an individual j had taken non-formal job-related education at least once in the previous 12 months at the time of the PIAAC survey. It takes a value of either one or zero representing participation or non-participation. Then, the effects of ICT skills and job-training participation on wages are captured in the parameters γ_{ICT} and γ_T , respectively.⁴

Work experience in (1) is often measured by years of potential labor market experience, which is typically defined as age or age

² This paper is a companion paper for Lee et al. (2020) which examine aging effects on productivity and a role of ICT capital investment in counterbalancing such aging effects using industry-level data. As a complement, this paper assesses such topics using individual-level data. While Lee et al. (2020) use value-added per labor as a measure of productivity, this paper considers wage as a measure of labor productivity at an individual level. In Korea, the observed wage, especially higher wage among older workers, must be determined not only by labor productivity, but also institutional features such as a seniority-based wage and promotion system and labor union.

³ We include a gender dummy in the wage equation to control for the wage differential by gender. When our specifications of (1), (2) and (3) without a gender dummy are applied to the sample of male workers, the main results remain robust.

⁴ Note again that this specification considers the effects of ICT skills and job-training participation on wages are equal to those on productivity. Using a large firm-level panel data set, Konings and Vanormelingen (2015) reported that the productivity increase from training is larger than the wage increase and argue that this gap can be explained by general training and imperfect competitions in the labor market.

minus year of schooling minus six. Potential labor market experience may be a good approximation. However, there is a discrepancy between potential and actual labor market experience. As this paper aims to estimate the “pure” age effect, we will include both age and actual labor market experience variables in the model as presented in 2(2) and (3). The age effect that we are interested in denotes the wage effect of age after controlling (actual) labor market experience and other explanatory variables, including gender, education attainment, ICT skills, and participation in job training. We will also consider the age structure of workers as another key factor for determining the effects of ICT skills and training on labor productivity. We will explore whether ICT skills and training can have relatively beneficial effects on the wages of the old workers aged 50–65 years, compared to the young workers.

We set up an extended version of the Mincer wage equation, which is our main specification, to examine the pure effects of aging as well as the heterogeneity effects of ICT skills and job training on labor productivity across age groups. We examine the interaction effects among ICT skills, job training, and the age of the workers. To elaborate, we are interested in whether ICT skills and training can have relatively beneficial effects on the wages of the older workers aged 50–65 years, compared to the younger workers. We extend the Mincer wage equation—(1)—by introducing the interaction term of age-group dummies with ICT skills and training, respectively.⁵

$$\log W_j = \alpha + \beta_G \text{Gender}_j + \beta_S \text{School}_j + \gamma_e \text{Job exp}_j + \gamma_a A_j + \gamma_{a_2} A_j^2 + \gamma_{ICT} \text{ICT skills}_j + \sum_{k=\text{middle, old}} \gamma_{ICT, \text{Age group}_k} (\text{ICT skills}_j \times \text{Age group}_k) + \gamma_T \text{Training}_j + \sum_{k=\text{middle, old}} \gamma_{T, \text{Age group}_k} (\text{Training}_j \times \text{Age group}_k) + \gamma_{LIT} \text{Literacy}_j + \varepsilon_j, \quad (2)$$

where $\log W_j$ is the logarithm of hourly earnings of individual j , School_j is a dummy for four educational attainment levels, which are lower secondary and below (HSD), upper secondary (HSG), college education (SME), and university and above (CLC). As mentioned earlier, instead of years of potential experience, we include Job exp_j in the empirical model, which is 2(2). It denotes the actual years of labor market experience that the individual j has, that is measured in years of paid work during his or her lifetime in the PIAAC survey. The model also has A_j as the age of individual j and its squared term to measure the pure aging effect, thereby the parameters γ_a and γ_{a_2} capture the effects of aging on wages of individuals.

In our model, we classify workers into three age groups: young (16–29 years), middle (30–49 years), and old (50–65 years). This model includes $\text{ICT skills}_j \times \text{Age group}_k$, interaction terms between a measure of ICT skills and two age-group dummies (middle and old), to explain how the wage effects of ICT skills differ by age groups, while considering young workers aged 16–29 as the reference group. Likewise, interaction terms ($\text{Training}_j \times \text{Age group}_k$) are added to identify differential training effects across age groups. In this empirical model, we are interested in the parameters γ_{ICT} and $\gamma_{ICT, \text{Age group}_k}$, which capture the relative wages of ICT skills on earnings across age groups, and the parameters γ_T and $\gamma_{T, \text{Age group}_k}$, which measure the relative training effects on earnings across age groups. In addition, workers unobserved ability could be still correlated with ICT skills so that able workers more likely to have high ICT skills. Thus, to control for endogeneity bias sourced from unobserved ability, we add a measure of literacy skills (Literacy_j) in the model.^{6,7}

Furthermore, we attempt to identify who will receive benefits from a higher level of ICT skills among the old by adding two different interaction terms to 2(2). One is between ICT skills, age-group dummies, and an indicator for receiving a tertiary level of education and the other is between ICT skills, age-group dummies, and an indicator for employment in a skill-intensive occupation. Then, we can construct Equation (3):

$$\begin{aligned} \log W_j = & \alpha + \beta_G \text{Gender}_j + \beta_S \text{School}_j + \gamma_e \text{Job exp}_j + \gamma_a A_j + \gamma_{a_2} A_j^2 + \gamma_{ICT} \text{ICT skills}_j + \sum_{k=\text{middle, old}} \gamma_{ICT, \text{Age group}_k} (\text{ICT skills}_j \\ & \times \text{Age group}_k) + \sum_{k=\text{middle, old}} \gamma_{ICT, \text{Age group}_k, \text{High edu}_j} (\text{ICT skills}_j \times \text{Age group}_k \times \text{High edu}_j) \\ & + \sum_{k=\text{middle, old}} \gamma_{ICT, \text{Age group}_k, \text{Skill occ}_j} (\text{ICT skills}_j \times \text{Age group}_k \times \text{Skill occ}_j) + \gamma_T \text{Training}_j \\ & + \sum_{k=\text{middle, old}} \gamma_{T, \text{Age group}_k} (\text{Training}_j \times \text{Age group}_k) + \varepsilon_j, \end{aligned} \quad (3)$$

where High edu_j is an indicator for receiving a high level of education, which refers to tertiary education (college and above), and Skill occ_j is an indicator for whether an individual j is employed in a skill-intensive occupation. Using this empirical model, we attempt to examine whether older workers who received a high level of education or are employed in a skill-intensive occupation may gain a relative benefit from the acquisition of ICT skills compared to their counterparts.

⁵ We performed a robustness test by adding industry fixed effects in our specifications, 2(2) and (3). The main results remain consistent. We will provide the results upon the request.

⁶ To address omitted variable bias, this paper uses literacy test scores of workers a proxy for the unobservable innate abilities. However, literary skills may not a perfect measure. Alternatively, we attempt to estimate Equations (2) and (3) by replacing literacy test scores with numeracy test scores and the results in Appendix Table 1 remain robust.

⁷ As shown in Fig. 5, literacy skills are not randomly distributed among workers and their scores are decreasing in age. We thank an anonymous reviewer for pointing out this. In this regard, we performed a robustness test by allowing heterogeneity in the innate abilities over age and cohorts. We added interaction terms between literacy skills and age-group dummies in Equations (2) and (3), and the estimation results in Appendix Table A2 remain consistent with our main results in Tables 2 and 3.

3. Data

To analyze the effects of aging on labor productivity, we use the Korean sample from the PIAAC survey. The PIAAC survey is an internationally comparable dataset developed by the OECD (2013a). It is based on the surveys of approximately 166,000 adults, aged 16–65 years, from 24 countries.⁸ The Korean sample consists of 6667 respondents, and provides detailed information including age, gender, education attainment, and job characteristics.

The PIAAC survey also provides detailed information related to ICT skills. There are three types of ICT skills-related information—skills proficiency in problem solving in technology-rich environments, use of ICT skills at work, and use of ICT skills at home. Using data on problem solving skills in technology-rich environments and the use of ICT skills in the workplace, we construct a composite measure of ICT skills. This measure considers not only an individual's ICT skills proficiency, but also takes into account his or her ability to use ICT skills in performing specific tasks at workplace, such as using e-mail, internet and spreadsheets.

In the previous studies on investigating the effects of ICT skills on wages or productivity considers problem-solving skills in technology-rich environment as a measure for ICT skills. However, it does not consider how such ICT skills are used in the workplace or whether an individual's ICT skills are adequately used in his or her job.

A notable feature of the PIAAC survey is to provide measures of adult skills proficiency in three domains—literacy, numeracy, and problem solving in technology-rich environments. The PIAAC defines problem solving in technology-rich environments as “using digital technology, communication tools and networks to acquire and evaluate information, communicate with others and perform practical tasks” (OECD, 2013a). The PIAAC survey focuses on problems that are closely related to the use of ICT and requires respondents to take the computer-based assessment. Hence, the respondent's proficiency score for problem solving in technology-rich environments is considered as a measure of his or her ICT skills attainment or proficiency. We will use normalized scores with a mean of zero and standard deviation of one.

However, there is a missing data problem. As reported in OECD (2013b), unlike literacy and numeracy proficiency assessments, proficiency scores for problem solving in technology-rich environments are only available for the respondents who took the computer-based assessment. For those who had never used a computer, those who refused to take the computer-based assessment and instead took the paper-and-pencil version, and those who failed the ICT core/basic assessment, the proficiency scores for ICT skills are not available. In total, in our sample, the proportion of these individuals without ICT skills proficiency scores is 25.9%. Unfortunately, those individuals with missing data are not random. They are most likely to have characteristics that are negatively correlated with their wage. For instance, those who had never used a computer are more likely to have a low level of educational attainment which is associated with a low wage.

Thus, we account for the missing data problem with an imputation method. In the estimation of imputation models, we use a regression imputation where literacy and numeracy test scores are adopted as main variables to predict the missing ICT skills proficiency measure. Literacy and numeracy test scores are good predictors in the regression model because we have literacy and numeracy scores for all individuals in the sample. Accordingly, the correlation between literacy and ICT skills proficiency and the correlation between numeracy and ICT skills proficiency are 0.81 and 0.77, respectively. We try various model specifications that differ in predictors by the degree of included individual characteristics (X_j) in the regression models such as:

$$ICT \text{ skills proficiency}_j = \alpha + \beta_1 \cdot \text{Literacy}_j + \beta_2 \cdot \text{Numeracy}_j + X_j \pi + u_j \quad 4$$

To avoid overfitting the problem of one-time imputation of the missing *ICT skills proficiency*_{*j*}, we introduce residual variance to predicted missing variables by using multiple imputation (MI) methods (Little & Rubin, 2002; Rubin, 1987). For instance, in the estimation of $\gamma_{ICT, Age \text{ group}_k}$ in Eq. (3), for each imputation we estimate $\hat{\gamma}_{ICT, Age \text{ group}_k, j}$ and we repeat this process *M* times (*j* = 1, 2, ..., *M*), and the MI estimate for $\gamma_{ICT, Age \text{ group}_k}$ is $\hat{\gamma}_{ICT, Age \text{ group}_k, MI} = \frac{1}{M} \sum_{j=1}^M \hat{\gamma}_{ICT, Age \text{ group}_k, j}$.⁹ In sum, we account for the sample selection issue due to missing data by applying multiple imputation methods while using occupation and industry fixed effects as well as the literacy variable to account for the omitted variable (e.g., unobserved individual ability) bias.

In addition, the PIAAC also reports the respondent's use of ICT skills in the workplace and in everyday life. Among the two measures, we decide to focus on the use of ICT skills in the workplace as our variable of interest in order to explore the wage effects of attaining ICT skills when a respondent is well-matched for their jobs or tasks on the basis of the ICT skills required for their jobs or tasks. According to OECD (2013a), “skill use at work can be defined as the level of skills that is observed in a worker's current job within a given skill domain. It is affected by both the extent to which workers are motivated to use their skills in the workplace and by the skills required to carry out the specific job”. The use of ICT skills in workplace, which is measured in an index, indicates how frequent and in what context an individual uses ICT skills. As noted in OECD (2013a), a higher value of the index corresponds to a more frequent use of ICT skills in his or her workplace.

To measure the use of ICT skills at work, the PIAAC asks two questions. It first asks whether the respondent does have experience with computer in work. It assigns a value of 0 to those with no experience with computer in their work, otherwise a value of 1. About 31.5% of the respondents reported that they have never performed tasks related to computer and/or internet in their work. The PIAAC

⁸ This is a cross-section dataset. Currently, the OECD is preparing for the second wave of the PIAAC survey. Once the second wave of the PIAAC survey is completed and published, we believe we can make more efforts to address endogeneity issues in measuring the effects of aging on labor productivity and the effects of ICT skills proficiency and job training on the labor productivity of older workers.

⁹ See Little and Rubin (2002) for more details and the formula to get the standard error for the estimate.

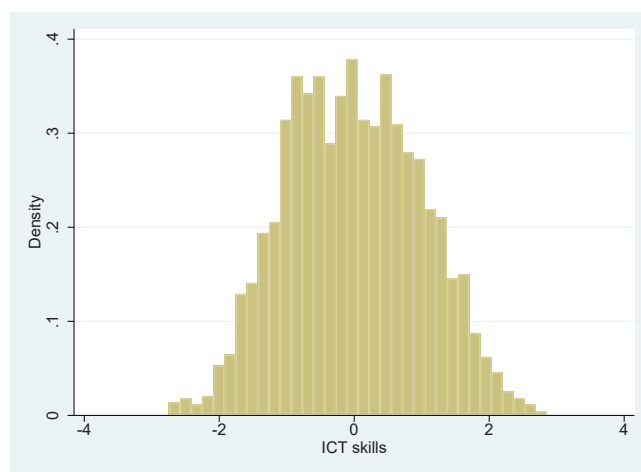


Fig. 4. Distribution of ICT skills. Notes: The measure of ICT skills is obtained by combining an individual's proficiency score in ICT skills and a score for their use at work. Specifically, it is constructed based on the summation of a standardized test score for problem-solving skill in a technology-rich environment and a standardized index for the use of ICT skills at work with an equal weight of 0.5. This combined score is then standardized again to have a mean of zero and a standard deviation of one.

Table 1
Descriptive Statistics of the Korean PIAAC data.

Variables	Full-time Korean workers	
	Mean	Standard deviation
ICT skills (Standardized composite index)	0	1
Basic Background		
Age (years)	40.0	11.1
16–29 group (share)	21.4	
30–49 group (share)	55.7	
50–65 group (share)	22.9	
Female (share)	42.2	
Hourly wage (Korean won)	12,992	8953
Educational Background (share)		
Low level of education		
Lower secondary and below (HSD)	12.0	
Upper secondary (HSG)	37.0	
High level of education		
Tertiary – College (SMC)	21.1	
Tertiary – University and above (CLC)	29.8	
Participation in non-formal job-related education	52.6	
Occupation/Jobs (share)		
Skill-intensive	68.0	
No. of Observations	2620	

Note: The sample consists of full-time Korean workers aged 16–65 who worked at least 30 h per week.

Source: PIAAC data from the OECD (2013a).

asks a follow-up question, to those having experience with computer in work, about the use of computers or internet as part of the respondent's work in the last job. The index of ICT skills use at workplace is obtained by encompassing information on “using e-mail, internet, spreadsheets, word processors, programming languages, and conducting transactions online, participating in online discussions (conferences, chats)” (OECD, 2013a). The value of this index ranges from 0.63 to 5.46 for those with experience with computer or internet. Finally, we standardize the index of ICT skills use with a mean of zero and a standard deviation of one.

We construct a measure of ICT skills using an individual's ICT skills proficiency and its use at work. First, we combine ICT skills proficiency, a standardized score of problem-solving skills in a technology-rich environment, and a standardized index for the use of ICT skills at work with an equal weight of 0.5. And then, we standardize this combined measure with a mean of zero and a standard deviation of one. OECD (2013b) notes that skill proficiency and skill use are mildly associated. This is also supported by a weak positive correlation between our sample's ICT skills proficiency score and ICT use index. Fig. 4 presents the distribution of our composite ICT skills measure having a mean of zero and standard deviation of one.

We restrict our sample to have only the full-time workers who worked more than 30 h per week. We exclude the self-employed from our sample, because their wages are not reported in the dataset. Then, our analysis is applied to 2620 Korean full-time workers.

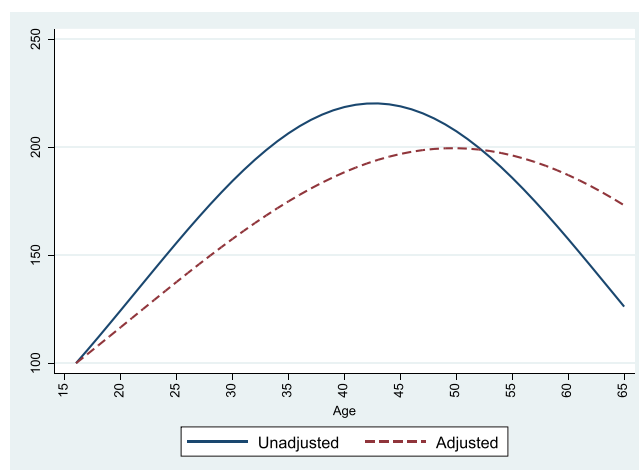


Fig. 5. Age and Wage Using the PIAAC Data. Notes: The unadjusted wage-age profile is drawn by using the coefficients for age and its squared term from the wage equation, without controlling for other variables, using the sample of the full-time workers from the PIAAC data. The adjusted one is constructed after controlling gender, educational levels, literacy skills and worker's job characteristics (i.e., occupation and industry).

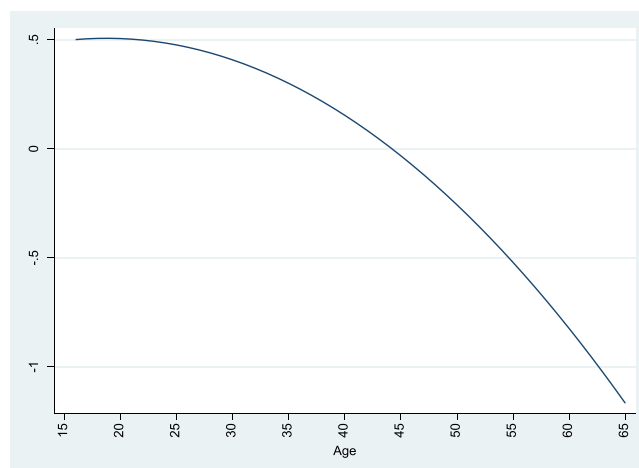


Fig. 6. Age and ICT Skills Using the PIAAC Data. Note: This figure is drawn based on the authors' calculation using the PIAAC sample.

Table 1 describes the characteristics of the Korean respondents in our sample. The Korean respondents were about 40.0 years old, on average. About 42.2% of them were female. The average hourly wage was estimated to be around 12,992 Korean won. We classify educational attainment of the respondents in the following four levels: lower secondary and below (HSD), upper secondary (HSG), college education (SME), and university and above (CLC). The respondents who completed HSG formed the largest group, amounting to 37.0%, followed by the graduates with CLC (29.8%) and those with college degrees (21.1%). The lowest share was of those with HSD (12.0%). About 52.6% of the respondents reported that they had participated in non-formal job-related education, that is job training, in the previous 12 months at the time of PIAAC survey. We construct additional education attainment variables by classifying them into two levels—low-educated (HSD and HSG) and highly educated (SME and CLC).

In addition, using the information on occupation measures based on skill level (ISCO-08), we classify the workers' occupation into two levels: skill-intensive and non-skill-intensive. A skill-intensive occupation refers to a skilled or semi-skilled white-collar occupation that requires an individual to conduct complex, technical, and practical tasks including decision-making with extensive professional knowledge. In contrast, a semi-skilled blue-collar or elementary occupation is considered as not a skill-intensive occupation, and the individual is required to perform simple, routine physical or manual tasks including operating machinery. In our sample, 68.0% of the full-time workers were employed in a skill-intensive occupation.

Fig. 5 illustrates two wage-age profiles using our individual-worker level data. The estimated wage-age profile, without controlling for other control variables—labeled as unadjusted wage-age profile in Fig. 5—is consistent with the previous studies supporting a hump-shaped relationship between wage and age. The unadjusted wage-age profile in Fig. 5 shows that wages increase until the worker reaches the age of around 42 and then it begins to rapidly decline. The decline at the later age may be attributable to the worker's education, job characteristics, and/or innate ability. On the other hand, after controlling the factors including the worker's

Table 2

Wage Equation with Aging, ICT skills and Job-training.

	(1)	(2)	(3)	(4)	(5)
Female	-0.216 * ** (0.022)	-0.215 * ** (0.022)	-0.214 * ** (0.022)	-0.211 * ** (0.022)	-0.210 * ** (0.022)
HSD	-0.133 * ** (0.048)	-0.120 * * (0.048)	-0.128 * ** (0.049)	-0.102 * * (0.049)	-0.107 * * (0.050)
SMC	0.187 * ** (0.029)	0.168 * ** (0.029)	0.168 * ** (0.029)	0.170 * ** (0.029)	0.170 * ** (0.029)
CLC	0.352 * ** (0.031)	0.327 * ** (0.031)	0.328 * ** (0.031)	0.327 * ** (0.031)	0.329 * ** (0.031)
Labor market experience	0.021 * ** (0.002)	0.020 * ** (0.002)	0.020 * ** (0.002)	0.020 * ** (0.002)	0.020 * ** (0.002)
Age	0.053 * ** (0.008)	0.053 * ** (0.008)	0.053 * ** (0.008)	0.049 * ** (0.009)	0.048 * ** (0.009)
Age ² /100	-0.062 * ** (0.009)	-0.062 * ** (0.009)	-0.062 * ** (0.009)	-0.059 * ** (0.011)	-0.058 * ** (0.011)
Indicator of ICT Skills Obs	-0.137 * ** (0.030)	-0.124 * ** (0.030)	-0.124 * ** (0.030)	-0.120 * ** (0.030)	-0.120 * ** (0.030)
ICT Skills	0.166 * ** (0.014)	0.155 * ** (0.014)	0.170 * ** (0.017)	0.141 * ** (0.023)	0.157 * ** (0.025)
ICT Skills x 30–49 Age group				0.017 (0.027)	0.018 (0.026)
ICT Skills x 50–65 Age group				0.029 (0.037)	0.039 (0.037)
Job Training		0.127 * ** (0.022)	0.127 * ** (0.022)	0.060 (0.041)	0.060 (0.041)
Job Training x 30–49 Age group				0.061 (0.046)	0.062 (0.045)
Job Training x 50–65 Age group				0.154 * * (0.062)	0.152 * * (0.062)
Literacy			-0.022 (0.018)		-0.025 (0.018)
Constant	7.950 * ** (0.147)	7.886 * ** (0.149)	7.886 * ** (0.148)	8.001 * ** (0.180)	8.013 * ** (0.180)
R-Square	0.391	0.397	0.397	0.399	0.400
Observations no.	2624	2620	2620	2620	2620

Notes: The dependent variable is the log of hourly earnings. The sample consists of full-time workers aged 16–65 who work at least 30 h per week. Robust Standard errors are in parentheses. The missing proportion is 25.9%. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

gender, education, job characteristics and literacy skills, productivity or wage does not show a steep decline at the later age, as can be seen in the adjusted wage-age profile in Fig. 5. Note that literacy skills are used as a proxy for the innate ability. In addition, attainment and utilization of ICT skills and participation in job training may further work against the decrease in a worker's productivity.

In addition, Fig. 6 describes the relationship between ICT skills and age. It shows that ICT skills also decrease with age. Older workers in the PIAAC data tend to have lower ICT skills compared to younger ones.

4. Wage effect of ICT skills and job training by age group

Table 2 presents estimation results of the Mincer wage equations—(1) and (2). Column (1) reports the estimates of the basic Mincer equation including ICT skills that incorporate an individual's ICT skills proficiency and his or her use of ICT skills at work. Column (2) adds an indicator for job training that takes a value of one if an individual had participated in job-specific, non-formal education within a year prior to the PIAAC survey. In Column (3), a score for literacy proficiency is included to account for unobserved abilities that can cause a bias in the estimation. Column (4) includes ICT skills and its interaction term with age-group dummies from 30 to 49 and 50–65 years as well as an indicator for job-training participation and its interaction term with the age-group dummies. Column (4) aims to assess whether ICT skills and job training can have relative beneficial effects on the wages of the old workers who are aged 50–65 years. In Column (5), we add literacy skills to Column (4).

In Column (1), the estimates on age and its squared variables are statistically significant and indicates that the effect of age on wages is positive until the worker reaches at the age of 42.7 and then turns negative afterward. Column (1) reports that the effect on the wages of ICT skills is positive and statistically significant. An increase in ICT skills by one standard deviation leads to a wage increase of 16.6% in Column (1).¹⁰ All other explanatory variables are consistent with those reported in the literature. The female indicator shows

¹⁰ Please note that in this specification, we do not identify whether an increase in ICT skills is caused by improvement in ICT skills proficiency of individual or his or her more frequent use of ICT skills in the workplace. We performed robustness tests by examining the wage effects of ICT skills proficiency and their use at work, respectively. The results are qualitatively similar to our main results in Tables 2 and 3. We will provide the results upon the request.

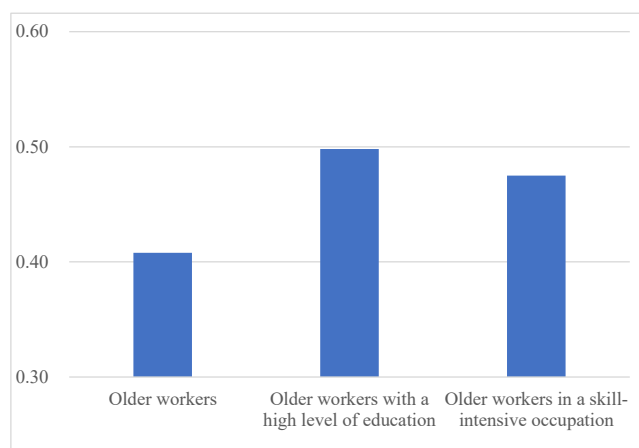


Fig. 7. Effect of Workers Aged 50–65 on Wages. Notes: The effects are drawn based on the estimates in Column (5) of Table 2 and Columns (1) and (2) of Table 3, respectively. The figures show the relative wage gains that the older workers aged 50–65 years earn when they obtain one-standard deviation above the mean of the ICT skills and participate in job training, compared to the workers with the mean value of ICT skills and no job training participation experience.

that female employees earn, on average, about 21.6% less than their male counterparts. The returns to schooling increase over the education levels and the reference group is high school graduates. An additional year of actual labor market experience leads to a wage increase of 2.1%.

Column (2) has the estimates of the basic Mincer wage regression with ICT skills and an indicator for job training participation. The estimate on ICT skills is positive and statistically significant. The estimated coefficient on job training of 0.127 indicates that workers who had participated in non-formal, job-related education during the previous year earn, on average, about 12.7% more than the non-participants with the same levels of age, labor market experience, education, and ICT skills.¹¹ Columns (1) and (2) provide supporting evidence that, on average, ICT skills and job-training participation can have beneficial effects on a worker's productivity.

However, the estimates on ICT skills and job training can be biased due to the endogeneity sourced from unobserved abilities. For instance, if workers with a higher level of innate ability are more likely to accept job training or ICT-related education, the effects of both ICT skills and job training would be overestimated. Column (3) includes proficiency scores for literacy skills as a proxy for the innate ability to control the unobserved innate ability that is closely correlated with ICT skills proficiency level as well as wages.¹² Column (3) shows that the results remain robust even after adding the literacy skills proficiency scores to Column (2). The wage effects of ICT skills and job training remain positive and statistically significant, respectively.

It is important to note that this study is more focused on investigating the relative wage effects that the older workers aged 50–65 years can have from ICT skills, rather than exploring the average wage effects of a worker from ICT skills. Thus, we restrain our attention on heterogeneous effects of ICT skills and job training by age. Then, we estimate 2(2) where ICT skills and job training variables interact with age groups.

Column (4) reports the estimates for ICT skills and its interactions with different age groups and those for an indicator for job training and its interactions with different age groups. Returns to ICT skills are positive and statistically significant. The estimated coefficient of 0.141 on ICT skill indicates that an increase in ICT skill by one standard deviation leads to a wage increase of 14.1% for a group of workers aged 15–29, which is the baseline age group. However, the estimates for the interaction term between ICT skills and age-group dummies turn out positive, but statistically insignificant, suggesting that the effects on the wages of ICT skills do not vary across age groups. Column (4) reports that the estimate for job training participation is positive but statistically insignificant, indicating that job-training participation does not have a significant wage effect for the group of workers aged 15–29. Also, the estimate for the interaction term between the job training participation indicator and a group of workers aged 30–49 is also positive and statistically insignificant. Interestingly, however, the estimate for the interaction term of job training participation with a group of workers aged 50–65 turns out positive and statistically significant. These results suggest that the wage effects of job training are greater for the workers aged 50–65. As shown in Column (4), for a worker in the age group of 50–65, the return to job training is 15.4% points greater

¹¹ The workers who received job-training much earlier than the previous 12 months at the time of survey (hereafter early receivers) were treated the same as those who had not received training. They belong to the control group. Given that receiving training affects wages positively, this early recipient of training could increase the comparison (or control) group's average wages as well. Then, the effect of job training can be underestimated as we cannot account for accumulative training due to data limitation. Furthermore, (under classical measurement error), measurement error causes attenuation bias, so that the effect of training may be under-estimated. Thus, our conclusion of a positive effect of training would not change due to measurement error.

¹² Although literacy is not a perfect measure for ability, given data limitation, it could be used to account for omitted innate abilities. The results remain the same regardless of using either literacy or numeracy test score as a proxy for an ability variable (see Appendix Table A1).

Table 3

Wage equation for aging, education, occupation, and ICT skills.

	(1)	(2)
Female	-0.208 *** (0.022)	-0.205 *** (0.022)
HSD	-0.126 ** (0.051)	-0.127 ** (0.050)
SMC	0.175 *** (0.029)	0.170 *** (0.029)
CLC	0.326 *** (0.031)	0.326 *** (0.030)
Labor market experience	0.020 *** (0.002)	0.020 *** (0.002)
Age	0.048 *** (0.009)	0.048 *** (0.009)
Age ² /100	-0.058 *** (0.012)	-0.058 *** (0.011)
Indicator of ICT Skills Obs	-0.126 *** (0.030)	-0.124 *** (0.030)
ICT Skills	0.153 *** (0.025)	0.153 *** (0.025)
ICT Skills x 30–49 Age group	-0.007 (0.034)	0.001 (0.036)
ICT Skills x 50–65 Age group	-0.004 (0.042)	-0.023 (0.042)
ICT Skills x 30–49 Age group x High Edu	0.038 (0.032)	
ICT Skills x 50–65 Age group x High Edu	0.137 ** (0.069)	
ICT Skills x 30–49 Age group x Skill Occ		0.021 (0.033)
ICT Skills x 50–65 Age group x Skill Occ		0.129 *** (0.044)
Job Training	0.063 (0.041)	0.061 (0.041)
Job Training x 30–49 Age group	0.062 (0.045)	0.065 (0.045)
Job Training x 50–65 Age group	0.149 ** (0.062)	0.155 ** (0.062)
Literacy	-0.019 (0.018)	-0.020 (0.019)
Constant	8.027 *** (0.183)	8.025 *** (0.181)
R-Square	0.401	0.402
Observations no.	2620	2620

Notes: The dependent variable is the log of hourly earnings. The sample consists of full-time workers aged 16–65 who work at least 30 h per week. Robust Standard errors are in parentheses. The missing proportion is 25.9%.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

than the reference—that is the young workers group aged 15–29. This indicates that when the workers who are aged 50–65 years have received job-related training at least once during the year, they obtain relative wage gains by 15.4% points compared to those who are aged 16–29 years and participated in job-related training.

Based on Column (5) of Table 2, we calculate relative productivity gains the older workers benefit from ICT skills and job training participation. For example, the older workers aged 50–65 who have one-standard deviation above the mean of the ICT skills and participate in job training earn 19.1% points ($\gamma_{ICT, Age\ group\ old} + \gamma_{T, Age\ group\ old} = 0.191$) more in wages than the younger workers aged 16–29, who possess one-standard deviation above the mean of the ICT skills and participate in job training. Fig. 7 presents the total wage effects that the older workers can earn, based on the estimates in Column (5) of Table 2. The total wage effect is estimated around 0.408 ($\gamma_{ICT} + \gamma_{ICT, Age\ group\ old} + \gamma_T + \gamma_{T, Age\ group\ old} = 0.408$). This suggests that when the older workers aged 50–65 obtain one standard deviation above the mean in their ICT skills and participate in job training, they can enjoy greater wage gains by 40.8% compared to the workers with a mean of ICT skills and no job training participation.

We also examine whether the effects of ICT skills on the productivity of older workers differs by education and skill levels. Therefore, we estimate Equation (3), which adds two distinct interaction terms—one between ICT skills, age-group dummies, and an indicator for having a high level of education, and the other between ICT skills, age-group dummies, and an indicator for employing in a skill-intensive occupation. By estimating this specification, we can identify who will receive greater benefits from higher level of ICT skills among the different subgroups of the older workers.

The estimation results are reported in Table 3. In Column (1), an interaction term between ICT skills, age-group dummies, and an indicator for receiving a high level of education are included. Column (2) has the interaction term between ICT skills, age-group

dummies, and an indicator for employing in a skill-intensive occupation. In all specifications, we also control literacy. As in the previous wage equations, most of the explanatory variables including age, gender, educational levels and labor market experience are statistically significant and their estimated parameters have the expected signs.

In Column (1), we add an interaction term between ICT skills, age-group dummies, and an indicator for an individual who received at least a college education. The estimated coefficient for ICT skills is 0.153, that is positive and statistically significant. This indicates that ICT skills has positive effect on workers' wage or productivity on the reference young-age group. In contrast, the estimate for the interaction term between ICT skills and the 50–65 age group dummy turns out negative, but statistically insignificant. More importantly, the estimated coefficient for the interaction term between ICT skills, the 50–65 age group, and an indicator for attaining a high level of education is estimated to be around 0.137, positive and statistically significant. It indicates that along with the one standard deviation increases in ICT skills, highly educated workers in the age range of 50–65 obtain relative wage gains by 13.7% points compared to low-educated workers in the same age group.

In Column (2), we add an interaction term between ICT skills, age-group dummies, and an indicator for whether an individual is employed in a skill-intensive occupation. Average return to ICT skills in the reference young-age group is positive and statistically significant. The estimate for the interaction term between ICT skills and the 50–65 age group dummy is negative and statistically insignificant. However, the coefficient for the interaction term between ICT skills, the 50–65 age group and an indicator for being employed in a skill-intensive occupation of 0.129 is positive and statistically significant. With the one standard deviation increase in ICT skills, the workers who are aged 50–65 years and are employed in a skill-intensive occupation obtain relative wage gains by 12.9% points compared to those in the same age group but holding a non-skill-intensive occupation.

Based on Columns (1) and (2) of Table 3, we calculate relative productivity gains of older workers aged 50–65 and received at least college-education or employed in a skill-intensive occupation to that of younger workers aged 15–29 years. The productivity increase of these groups of older workers with at least college-education is significantly positive when they obtain one-standard-deviation above the mean of the ICT skills and participate in job training. The estimated effect of 0.282 ($\gamma_{ICT, Age\ group\ old} + \gamma_{ICT, Age\ group\ old, High} + \gamma_{T, Age\ group\ old} = 0.282$) in column (1) indicates that they benefit greater productivity increase by 28.2% points, compared to younger workers, from acquiring and utilizing a higher level of ICT skills and participating in job-training programs. Productivity gains for the older workers in a skill-intensive occupation are also estimated in column (1). These older workers have greater productivity gains by 26.1% points, relative to that of younger workers ($\gamma_{ICT, Age\ group\ old} + \gamma_{ICT, Age\ group\ old, Skill\ occ} + \gamma_{T, Age\ group\ old} = 0.261$), when they obtain one-standard-deviation above the mean of the ICT skills and participate in job training.

In addition, Fig. 7 presents the total effects of ICT skills and job-training participation for the older workers in the age group of 50–65. The one with at least a college education and the other one with a skill-intensive occupation. Based on the estimates in Column (1) of Table 3, for the older workers who completed at least college education (denoted as high-educated), the total wage effect is calculated around 0.498 ($\gamma_{ICT} + \gamma_{ICT, Age\ group\ old} + \gamma_{ICT, Age\ group\ old, High} + \gamma_T + \gamma_{T, Age\ group\ old} = 0.498$). This suggests that the highly educated older workers, who own one-standard-deviation above the mean of the ICT skills and participate in job training, enjoy greater wage gains by 49.8% compared to the workers with the mean value of ICT skills and no job training experience. Similarly, we also calculate the total wage effect for the older workers employed in a skill-intensive occupation, using the estimates in Column (2) of Table 3. The total wage effect of 0.475 ($\gamma_{ICT} + \gamma_{ICT, Age\ group\ old} + \gamma_{ICT, Age\ group\ old, Skill\ occ} + \gamma_T + \gamma_{T, Age\ group\ old} = 0.475$) indicates that the older workers employed in a skill-intensive occupation having one-standard deviation above the mean of the ICT skills and participating in job training obtain greater wage gains by 47.5% compared to the workers with the mean value of ICT skills and no job training participation experience.

Hence, the results in Table 3 identify who can gain greater benefits from a higher level of ICT skills among the older workers. Older high-skilled workers, such as those who are well-educated or hold a skill-intensive occupation can enjoy relative wage gains from of a higher level of ICT skills, compared to older low-skilled workers.

5. Discussion

In this section, we discuss how we have attempted to address the sources of endogeneity bias sourced from omitted variable bias, missing data problems, and self-selection in estimating the Mincer-type equations. As previously mentioned, we attempt to control for omitted variable bias. As for the omitted variable bias, we assume that there is an innate ability that may affect ICT skills and worker's wages and use literacy scores as a measure for the innate ability when estimating 1(1) through (3).

Another major endogeneity bias may come from missing data for ICT skills proficiency. As already mentioned, ICT skills measure are available only for those individuals who took the computer-based assessment. In our sample, the proportion of the individuals without test scores for ICT skills proficiency accounts for 25.3%. The individuals with missing data are not random; thus, we account for the missing data problem with an imputation method. In the estimation of imputation models, we use a regression imputation where literacy and numeracy test scores are adopted as the main variables to predict missing ICT skills measure. In addition to the imputation method, following Busso, DiNardo, and McCrary (2014), we also perform a test by adopting an inverse-probability weighting (IPW) to see whether controlling for the missing data problem in the case of ICT skills proficiency is appropriate. This is a commonly used method, and it is used to adjust for unequal sampling fractions in sample surveys. We first use a logit model to predict the probability to take the computer-based assessment and have ICT skills proficiency scores in the PIAAC survey using the respondent's individual characteristics and background variables (gender, education levels, work experience, mother's education, participation in job training, literacy, industry, and occupation). Accordingly, we use this probability as a (inverse) weight in estimating the Mincer equation. The estimation results using an IPW provide supporting evidence for our main results that the wage effects

of ICT skills and job trainings are positive and greater for workers aged 50–65 if they are at least college-educated or employed in a skill-intensive occupation.¹³

In line with the missing data problem, the other major source of endogeneity bias is self-selection in achieving a high level of ICT skills and/or participating in job training. As previously mentioned, ICT skills and participation in job training are not randomly determined. For instance, an individual who expects to have a higher wage through ICT skills and participation in job-related training is more likely to attain ICT skills, perform ICT-related tasks, and/or join job-related training courses. To deal with this self-selection issue, we have attempted several econometric techniques including instrument variable methodology, and difference-in-difference.

We find that it is extremely difficult to come up with appropriate instruments for ICT skills and/or job training. Falck et al. (2021) who estimate the returns to ICT skills across 19 countries and across German regions use (regional) variation in the extent of the traditional voice-telephony network, which existed before the introduction of broadband across countries and German municipalities as an instrument for ICT skills. This instrument variable may not be appropriate in the case of Korea where the roll-out of broadband network is not determined by the traditional voice-telephony network but by the government intervention. A possible instrument is the presence of computer at home during childhood, but this data is not available in the PIAAC survey.

As mentioned in the Introduction, the implementation of “Computer” courses at schools over the period, 2001–2005 is one of the Korean government’s efforts to increase ICT literacy and familiarize students and people with ICTs. We estimated the effect of ICT skills on wages using the policy intervention of “Computer” courses at schools during 2001–2005, which allows using cohorts just before as control group and cohorts just after as treatment group. We applied average treatment effects analysis and found no significant difference between these two groups for wages’ ICT skills. Because of the lack of observations around the cutoff, we do not have enough statistical power to find any discernable difference between these two groups in ICT skills. Hence, instead, we attempt to control observable variables using the propensity score and estimate ICT skills’ role on wages across age cohorts—unfortunately, the results are somewhat sensitive.

As for participation in job training, distance to job training center and availability of training facilities might be a possible instrument for participation in job training. As noted in Lee et al. (2019), regional dummies for availability of training facilities turn out to be a weak instrument, and the results are not quantitatively sensitive to the endogeneity issue. We leave thorough investigation of the endogeneity issues for future studies.

6. Conclusion

This paper assessed the effects of aging on labor productivity, using the individual worker-level data. We found that ICT skills can have significantly positive effects on the productivity of older workers who are highly educated or employed in skill-intensive occupations. To be specific, the estimation results showed that the wage effects of ICT skills are greater for those with a high level of education or in a skill-intensive occupation among older workers. We also found the wage effects of job training are greater for the workers aged 50–65 than any other age groups. The estimation results also highlight that compared to younger workers, the older skilled workers can benefit significantly more from attaining and utilizing ICT skills and job training participation. These results suggest that productivity decline due to the aging process can be mitigated by promoting training for older workers to equip them with adequate ICT skills, even with a larger productivity than the younger workers.

Our findings suggest that Korea’s rapid demographic shift toward an aged society is not a mere threat to its economy but offers an opportunity to utilize more-mature workers’ skills in productive ways with the support of technologies and lifelong education. Firms’ investment in ICT technologies and training of ICT skills can help improve productivity and extend the working age of aging populations.

Our empirical investigations are subject to several limitations. As previously discussed, we have attempted to address the major endogeneity issues using different empirical techniques such as inclusion of literacy skills scores, and fixed-effects, and found that the results were robust. However, these techniques may not fully control for the potential endogeneity problems. The use of valid instrumental variables and the implementation of experimental settings to control for endogeneity in the ICT skills and job training participation would provide consistent estimates, but the ideal instruments are hard to find. We leave this issue for future studies.

Acknowledgements

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Appendix

See Appendix Tables A1 and A2.

¹³ We will provide the estimation results upon the request.

Table A1
Wage equation with numeracy test score.

	(1)	(2)	(3)	
Female	-0.215 * ** (0.022)	-0.211 * ** (0.022)	-0.209 * ** (0.022)	-0.206 * ** (0.022)
HSD	-0.124 * * (0.049)	-0.105 * * (0.050)	-0.125 * * (0.051)	-0.125 * * (0.050)
SMC	0.168 * ** (0.029)	0.170 * ** (0.029)	0.176 * ** (0.029)	0.170 * ** (0.029)
CLC	0.328 * ** (0.031)	0.329 * ** (0.031)	0.326 * ** (0.031)	0.326 * ** (0.031)
Labor market experience	0.020 * ** (0.002)	0.020 * ** (0.002)	0.020 * ** (0.002)	0.020 * ** (0.002)
Age	0.053 * ** (0.008)	0.049 * ** (0.009)	0.048 * ** (0.009)	0.048 * ** (0.009)
Age ² /100	-0.062 * ** (0.009)	-0.059 * ** (0.011)	-0.059 * ** (0.012)	-0.058 * ** (0.011)
Indicator of ICT Skills Obs	-0.128 * ** (0.032)	-0.125 * ** (0.031)	-0.129 * ** (0.032)	-0.127 * ** (0.031)
ICT Skills	0.162 * ** (0.016)	0.149 * ** (0.025)	0.146 * ** (0.025)	0.146 * ** (0.025)
ICT Skills x 30–49 Age group		0.017 (0.027)	-0.010 (0.034)	-0.002 (0.036)
ICT Skills x 50–65 Age group		0.034 (0.037)	-0.010 (0.042)	-0.029 (0.042)
ICT Skills x 30–49 Age group x High Edu			0.041 (0.032)	
ICT Skills x 50–65 Age group x High Edu			0.142 * * (0.069)	
ICT Skills x 30–49 Age group x Skill Occ				0.025 (0.033)
ICT Skills x 50–65 Age group x Skill Occ				0.131 * ** (0.044)
Job Training	0.126 * ** (0.022)	0.060 (0.041)	0.063 (0.041)	0.061 (0.041)
Job Training x 30–49 Age group		0.061 (0.045)	0.061 (0.045)	0.064 (0.046)
Job Training x 50–65 Age group		0.154 * * (0.062)	0.150 * * (0.062)	0.156 * * (0.062)
Numeracy	-0.011 (0.018)	-0.014 (0.018)	-0.008 (0.018)	-0.010 (0.018)
Constant	7.882 * ** (0.148)	8.000 * ** (0.180)	8.020 * ** (0.183)	8.017 * ** (0.181)
R-Square	0.397	0.399	0.401	0.402
Observations no.	2620	2620	2620	2620

Notes: The dependent variable is the log of hourly earnings. The sample consists of full-time workers aged 16–65 who work at least 30 h per week. To account for an omitted ability variable, numeracy test scores are used. Columns (1) and (2) of Appendix Table A1 correspond to Columns (3) and (5) of Table 2 and Columns (3) and (4) correspond to Columns (1) and (2) of Table 3, respectively. Robust Standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A2

Wage Equation with Literacy Test Score and its Interaction Terms with Age-groups.

	(1)	(2)	(3)	(4)
Female	-0.212 * ** (0.022)	-0.210 * ** (0.022)	-0.209 * ** (0.022)	-0.205 * ** (0.022)
HSD	-0.137 * ** (0.052)	-0.117 * * (0.051)	-0.130 * * (0.052)	-0.134 * ** (0.051)
SMC	0.166 * ** (0.029)	0.164 * ** (0.029)	0.169 * ** (0.029)	0.165 * ** (0.029)
CLC	0.325 * ** (0.031)	0.320 * ** (0.031)	0.317 * ** (0.031)	0.318 * ** (0.031)
Labor market experience	0.020 * ** (0.002)	0.020 * ** (0.002)	0.020 * ** (0.002)	0.020 * ** (0.002)
Age	0.052 * ** (0.008)	0.046 * ** (0.009)	0.044 * ** (0.010)	0.045 * ** (0.010)
Age ² /100	-0.061 * ** (0.009)	-0.054 * ** (0.012)	-0.053 * ** (0.012)	-0.053 * ** (0.012)
Indicator of ICT Skills Obs	-0.126 * ** (0.030)	-0.123 * ** (0.030)	-0.127 * ** (0.030)	-0.126 * ** (0.030)
ICT Skills	0.169 * ** (0.017)	0.158 * ** (0.031)	0.159 * ** (0.031)	0.158 * ** (0.031)
ICT Skills x 30–49 Age group		-0.006 (0.037)	-0.036 (0.044)	-0.032 (0.047)
ICT Skills x 50–65 Age group		0.175 * ** (0.067)	0.132 * (0.070)	0.103 (0.075)
ICT Skills x 30–49 Age group x High Edu			0.043 (0.032)	
ICT Skills x 50–65 Age group x High Edu			0.084 (0.070)	
ICT Skills x 30–49 Age group x Skill Occ				0.030 (0.034)
ICT Skills x 50–65 Age group x Skill Occ				0.106 * * (0.045)
Job Training	0.125 * ** (0.022)	0.063 (0.041)	0.067 * (0.041)	0.065 (0.041)
Job Training x 30–49 Age group		0.052 (0.045)	0.050 (0.045)	0.054 (0.045)
Job Training x 50–65 Age group		0.150 * * (0.062)	0.147 * * (0.062)	0.152 * * (0.062)
Literacy	-0.044 (0.030)	-0.024 (0.039)	-0.024 (0.039)	-0.025 (0.039)
Literacy x 30–49 Age group	0.041 (0.029)	0.038 (0.044)	0.043 (0.044)	0.042 (0.044)
Literacy x 50–65 Age group	0.006 (0.038)	-0.129 * (0.067)	-0.114 * (0.067)	-0.109 (0.068)
Constant	7.915 * ** (0.152)	8.053 * ** (0.182)	8.073 * ** (0.186)	8.068 * ** (0.184)
R-Square	0.398	0.403	0.404	0.404
Observations no.	2620	2620	2620	2620

Notes: The dependent variable is the log of hourly earnings. The sample consists of full-time workers aged 16–65 who work at least 30 h per week. To account for an omitted ability variable, literacy test scores and its interaction terms with age groups are used. Columns (1) and (2) of Appendix Table A2 correspond to Columns (3) and (5) of Table 2 and Columns (3) and (4) of correspond to Columns (1) and (2) of Table 3, respectively. Robust Standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

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